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Assessing the influence of generative artificial intelligence (GenAI) on awareness and behavior in medical research integrity: An online survey study

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ABSTRACT

Background: Generative Artificial Intelligence(GenAI) significantly enhances medical research efficiency but raises ethical concerns regarding research integrity. The lack of systematic guidelines for its ethical use underscores the need to investigate GenAI's impact on researchers' awareness and behavior concerning integrity.

Methods/Materials: A cross-sectional survey of 718 valid responses from Chinese medical researchers assessed GenAI's impact on research integrity using an extended Unified Theory of Acceptance and Use of Technology(UTAUT) model.

Results: The findings reveal that performance expectancy, effort expectancy, technical environment, trust in technology, and supporting conditions positively influence researchers' awareness of research integrity. Conversely, GenAI anxiety and perceived risks exert a significant negative impact. Furthermore, both supporting conditions and integrity awareness are positively associated with integrity behavior, while GenAI anxiety negatively affects such behavior.

Conclusion: The stakeholders in the medical research ecosystem should develop comprehensive guidelines for the responsible use of GenAI. Emphasis should be placed on optimizing the technical environment, enhancing trust and support structures, and embedding integrity safeguards, thereby promoting the synergistic development of technological innovation and ethical research practices.

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KEYWORDS

Generative AI; research integrity; UTAUT model; technical trust; perceived risk anxiety

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1. Introduction

1.1. *Social background: Emerging challenges to medical research integrity*

Compared to other disciplines, medical research demands higher ethical standards and carries greater social responsibility, as it is directly tied to human life and health (F. Wang, Hou, and Zhang 2025). Any breach of integrity in this field can lead to misleading medical conclusions – undermining academic credibility and, more critically, posing significant risks to patient safety (Haitao, Xufeng, and Wen 2025). In recent years, there has been a noticeable global increase in the retraction of medical papers, with several hospitals and medical schools becoming embroiled in misconduct scandals that have triggered crises of public trust (Hua 2025; Noorden 2025). Thus, the integrity of medical research extends beyond compliance with academic norms (Ivan David et al. 2025); it represents a fundamental pillar for safeguarding public health, maintaining scientific credibility, and advancing the quality of healthcare.

1.2. *Technical background: GenAI is rapidly penetrating the entire medical research lifecycle*

GenAI employs neural networks and deep learning algorithms to produce coherent and logically structured content across multiple modalities – including audio, code, images, text, simulations, and video – based on the data it has been trained on (Company 2024). Within the realm of medical research, GenAI exhibits considerable potential, particularly in text generation and data processing (Jérémie and David 2024), offering novel methodological tools that can significantly enhance research efficiency and innovation (Alisha et al. 2024; Tanatorn et al. 2024; Xufei et al. 2025). Prior studies have employed GenAI to enhance the accuracy of medical image analysis and assisted adaptive cancer therapy (Tanatorn et al. 2025; Vishwanatha et al. 2025; Youcef 2025).

Currently, GenAI is being rapidly adopted across a wide array of sectors, including education (Y. Guoming, Li, and Teng 2024; H. Li 2023), journalism and media (Y. U. Guoming and Jianwei 2023), visual arts (Xiong 2017), finance (Dwivedi et al. 2023), and hospitality (Dwivedi et al. 2024), fundamentally transforming traditional models of knowledge production and dissemination. Studies have shown that GenAI fosters pedagogical innovation in higher education (Chiu et al. 2023), and improves the efficiency and impact of media and publishing (Lu 2023). It has received widespread acclaim from leading authorities such as *MIT Technology Review*, Gartner, and *Nature* (Groombridge et al. 2021, Nature 2023). Nevertheless, the powerful generative capabilities of GenAI have also sparked growing concerns, including issues related to data privacy, disinformation, algorithmic

and racial biases, ethical risks, and heightened levels of AI-induced anxiety (Bauchner, Steinbrook, and Redberg 2025; Monteith et al. 2024; Tang 2023; Xie and Li 2023).

As GenAI increasingly integrates into all phases of medical research, it becomes imperative to investigate its influence on the integrity of scientific practices (Kayte 2025). In particular, there is a growing need to understand how GenAI may affect medical researchers' cognitive judgments and behavioral choices related to research integrity. This study employs an empirical research approach to investigate the proposed questions in a systematic and data-driven manner.

While constructs such as performance expectancy, effort expectancy, facilitating conditions, and social influence have been extensively examined and validated within the unified theory of acceptance and use of technology (UTAUT), the emergence of technologies like GenAI introduces additional factors that may influence medical researchers' behavioral intentions. Accordingly, this study extends the UTAUT framework by incorporating the dimensions of technical trust, GenAI-anxiety and GenAI-perceived risk. Technical trust refers to the extent to which an individual perceives a given technology as trustworthy, reliable, and secure (Chen, Fan, and Azam 2024). GenAI-Perceived risk is the expectation of potential problems following the introduction of GenAI technology (M. C. Lee 2009). GenAI-Anxiety refers to apprehension and fear of using new and unfamiliar GenAI technologies (Budhathoki et al. 2024). This study aims to systematically investigate the attitudes and adoption mechanisms of medical researchers toward GenAI in the context of research integrity. It seeks to deepen the understanding of how GenAI influences researchers' awareness and behaviors related to research integrity, thereby contributing to the sustainable development of this emerging field. By employing structural equation modeling (SEM), this study incorporates constructs such as performance expectations, effort expectation, supporting conditions, technical environment, trust, anxiety, and perceived risk into the research framework. The findings are expected to offer novel theoretical insights and practical implications for promoting research integrity.

The following chapters are organized as follows: we first review the existing literature on GenAI and medical research integrity, followed by a detailed description of the research methodology and data analysis procedures. We then present and discuss the empirical findings, and conclude with a summary of the key results and implications, along with suggestions for future research. Through this work, we aim to enhance the understanding of medical researchers' attitudes toward GenAI-assisted research and their intentions to engage in integrity behaviors. This study provides both a theoretical foundation and practical guidance for the responsible integration of GenAI into the medical research landscape. Moreover, by

incorporating the constructs of technical trust, GenAI-anxiety, and GenAI-perceived risk into the research framework, we offer actionable recommendations for medical institutions seeking to improve and optimize the ethical and effective use of GenAI technologies.

2. Theoretical support and research hypotheses

2.1. Variables under the unified theory of acceptance and use of technology (UTAUT)

The Unified Theory of Acceptance and Use of Technology (UTAUT), proposed by Venkatesh(2003) (Venkatesh et al. 2003), is a comprehensive framework developed by integrating several prominent models of technology acceptance. These include the Theory of Reasoned Action, the Technology Acceptance Model, the Motivational Model, the Theory of Planned Behavior, the Innovation Diffusion Theory, and Social Cognitive Theory (Williams, Rana, and Dwivedi 2015). The UTAUT model (Figure 1) is designed to predict user acceptance and use of technology by accounting for multiple influencing factors and individual differences. It identifies four core constructs – performance expectancy, effort expectancy, social influence, and facilitating conditions – along with four moderating variables: gender, age, experience, and voluntariness of use (Venkatesh, Thong, and Xu 2016).

UTAUT has been widely employed by researchers both domestically and internationally to investigate user acceptance and behavior across various technological contexts. For example, performance expectancy and effort expectancy significantly influenced users’ willingness to adopt chatbot services (Balakrishnan, Abed, and Jones 2022), while social influence and facilitating conditions either promoted or hindered the adoption of GenAI in new product development teams (Xia and Chen 2024). Some studies have also applied the UTAUT model to evaluate the adoption of ChatGPT in

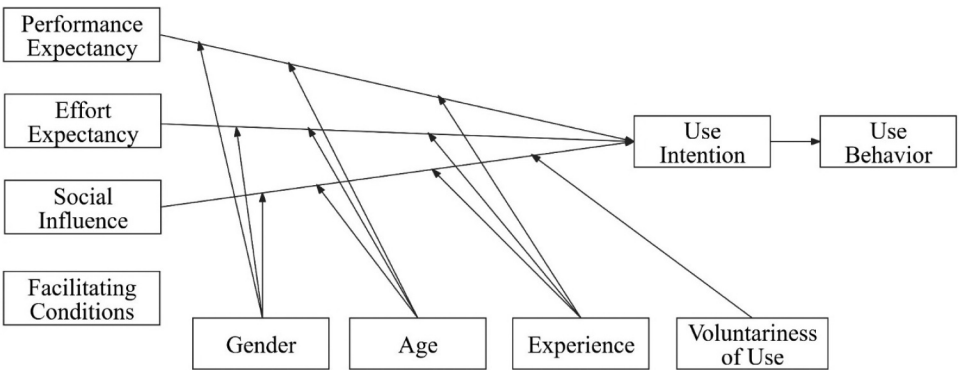


Figure 1. UTAUT model.

medical education (Noura et al. 2025), understanding older adults' intention to use telehealth (Anne, Joneen, and Kristen 2025), and examine students' and teachers' acceptance and use of educational technologies (Xue, Rashid, and Sha 2024).

2.1.1. Performance expectations (PE)

In this study, performance expectations refer to medical researchers' subjective perceptions of the extent to which GenAI can enhance research efficiency and improve the quality of research outputs. Research shows that performance expectations significantly influence users' evaluation of generative artificial intelligence (Jiayu et al. 2025; Qingran 2025). When researchers perceive GenAI as a valuable tool for optimizing research workflows and producing higher-quality outcomes, they are more likely to adopt the technology in a responsible and compliant manner. This responsible use, in turn, may reinforce their awareness of research integrity. Based on this, the following hypotheses are proposed:

H1: In medical research activities, higher performance expectations regarding GenAI are significantly associated with increased awareness of research integrity among medical researchers.

2.1.2. Effort expectation (EE)

In this study, effort expectation refers to medical researchers' perceptions of the operational ease, learning cost, and usability burden associated with GenAI technologies. It reflects their subjective assessment of how easy GenAI is to learn and use in the context of research practice. When researchers perceive GenAI as intuitive and easy to integrate into their workflows, they are more inclined to invest effort in understanding its responsible application and to engage more consciously with research integrity norms – viewing them as essential to ensuring research credibility. Conversely, if GenAI is perceived as complex, time-consuming, or difficult to navigate, researchers may prioritize convenience over compliance, potentially overlooking ethical requirements and increasing the likelihood of misconduct. Research shows that effort expectancy significantly positively affect behavioral intention (Jiayu et al. 2025). Thus, greater perceived ease of use is expected to enhance researchers' awareness of research integrity when adopting GenAI. Based on this, the following hypotheses are proposed:

H2: In medical research activities, a higher level of effort expectation is significantly associated with greater awareness of research integrity among medical researchers.

2.1.3. Technical environment (TE)

In the context of medical research, the broader technical environment encompasses national regulatory frameworks, editorial policies from academic publishers, and institutional management systems that collectively shape the compliant use of GenAI technologies. Unlike the broader construct of “social influence” the technical environment more precisely captures the external structural factors that directly govern how medical researchers utilize GenAI in practice. Social influence significantly affect learners’ behavioral intention to use GenAI (Qingran 2025). For example, national policies define the overarching regulatory boundaries for AI usage; journal publication guidelines dictate how AI assistance must be disclosed in manuscripts; and internal rules within medical institutions regulate the permissible use of GenAI in scientific workflows.

This study defines the technical environment as the constellation of external support systems – such as policy directives, publication standards, and institutional protocols – that guide researchers in the ethical and appropriate integration of GenAI. A clearly articulated regulatory infrastructure can reduce ambiguity, prevent misuse, and reinforce adherence to research integrity norms. In contrast, vague or inconsistent external guidance may lead to confusion, potentially fostering misconduct such as plagiarism, improper attribution, or concealment of AI involvement. Therefore, a well-structured technical environment serves as a critical external condition for enhancing researchers’ awareness of research integrity. Based on this, the following hypotheses are proposed:

H3: In medical research activities, a well-established technical environment is significantly associated with greater awareness of research integrity among medical researchers.

2.1.4. Supporting conditions (SC)

In medical research contexts, technology adoption by researchers is influenced not only by individual perceptions but also by objective factors such as hardware infrastructure, user interfaces, and network environments. To more accurately capture these real-world constraints, this study replaces UTAUT’s original “facilitating conditions” with the concept of supporting conditions. We define supporting conditions as the availability of comprehensive

infrastructure and technical support that directly enable researchers to use GenAI. Study shows that facilitating conditions positively influence the behavioral intention to use GenAI tools (Dominika, Cintia, and Ana Isabel 2025).

When supporting conditions are robust, researchers face lower operational burdens and are more willing to adopt GenAI in a compliant manner, thereby fostering greater awareness of research integrity. Moreover, adequate technical support ensures that researchers can execute workflows according to established norms, reducing the likelihood of misconduct arising from technological obstacles and enhancing their actual integrity behaviors. Based on this, the following hypotheses are proposed:

H6: In medical research activities, strong supporting conditions are significantly associated with higher levels of research integrity awareness among medical researchers.

H10: In medical research activities, strong supporting conditions are significantly associated with improved research integrity behavior among medical researchers.

2.2. Technical trust, GenAI-Anxiety and GenAI-Perceived risk

GenAI built upon large language models (LLMs) and deep learning technologies, possesses powerful content generation capabilities and is rapidly reshaping the entire medical research lifecycle. Prior studies have demonstrated that GenAI significantly influences multiple stages of the scientific research process, including data collection, statistical analysis, figure generation, manuscript writing, language editing, citation management, authorship attribution, peer review, and even manuscript submission and publication workflows (Cooper 2023; P. Wang and Xu 2023). Specifically, GenAI can support researchers in creative ideation (Bouschery, Blazevic, and Piller 2023), scientific writing (Silva et al. 2023), and the organization and analysis of research data (Blandchand et al. 2023), substantially improving research productivity and the overall quality of scientific output.

The widespread integration of GenAI into academic research has prompted swift policy responses from major international academic publishers. Leading journals such as *Science*, *The Lancet*, and *Cell* have publicly stated that ChatGPT and similar AI tools cannot be credited as authors, though their use as research assistants is permitted. Key regulatory bodies – including the International Committee of Medical Journal Editors (ICMJE), the World Association of Medical Editors (WAME), and the Council of

Science Editors (CSE) – have issued formal guidelines for the responsible use of AI-generated content. For instance, *Nature* has published specific editorial recommendations on the application of ChatGPT in academic writing and peer review, aiming to define ethical boundaries and acceptable usage practices (Gruda 2024). Current literature indicates that GenAI has a dual impact on the integrity of medical research, offering both potential benefits and emerging risks. On the positive side, GenAI can facilitate paradigm shifts in research methodologies (Ruksakulpiwat, Kumar, and Ajibade 2023), assist in disease data screening (Cao and Liu 2022), and enhance research efficiency (P. Y. Lee et al. 2023). Conversely, concerns have been raised about reduced credibility in AI-assisted research findings (Saini and Saxena 2023), compromised publication integrity (Rahimi and Abadi 2023), ambiguous intellectual property ownership (Jeyaraman et al. 2023), and the absence of well-established ethical norms.

As the technology continues to evolve, domain-specific GenAI models tailored for medical applications have emerged. Examples include BioMedLM 2.7B, LLaMA (Large Language Model Meta AI), Bio-BERT, Bio-NLP, Med-PaLM 2, Bio-Megatron, Clinical-BERT, XGen-7B, GatorTronGPT, Chat-Doctor, and others (Cascella et al. 2024; H. Guo et al. 2024). These models demonstrate specialized capabilities such as comprehension of medical terminology, integration of interdisciplinary knowledge, and interpretation of clinical language. They have been applied to a wide range of tasks, including dental research (Liwen and Yang 2024), drug development pipelines (Y. Li 2023), mining of drug – disease associations (Gao et al. 2024), and optimization of cancer screening and treatment protocols (Iannantuono et al. 2023). Moreover, GenAI is increasingly used to support clinicians by providing multidimensional information, aiding clinical decision-making, and enhancing diagnostic and therapeutic strategies (Thirunavukarasu et al. 2023; Walker et al. 2023; F. Wang 2023). In areas such as pathology and clinical imaging, AI systems have shown high levels of accuracy and reliability in image recognition, automated diagnostics, and report generation (Rau et al. 2023; Schukow et al. 2024). These technologies can also help patients better understand test results, fostering greater trust and engagement in medical decision-making processes (Cadamuro et al. 2023).

2.2.1. GenAI-Perceived risk (GenAI-PR)

Perceived risk has been widely studied in areas such as e-commerce (Lin et al. 2025), sports lottery purchasing behavior (Özkan A, Begum, and Murat 2025), and acceptance of GenAI (Sirui and Yuhong 2025). In this study, GenAI-perceived risk refers to the subjective assessment by medical researchers of the potential threats and negative consequences associated with using GenAI in scientific research activities. These perceived risks include concerns

over data privacy breaches, the dissemination of misinformation, violations of ethical norms, and the possible misuse or misinterpretation of AI-generated outputs. When medical researchers perceive high levels of uncertainty or ethical risk associated with GenAI, they may – whether consciously or unconsciously – engage in nonstandard research behaviors in an attempt to avoid potential pitfalls or to secure desired outcomes. This may include actions such as concealing AI assistance, bypassing review protocols, or downplaying the limitations of AI-generated data. Over time, such practices can erode researchers' commitment to upholding scientific integrity. Moreover, heightened risk perception may foster distrust toward GenAI technologies, resulting in anxiety about proper usage and diminishing adherence to research ethics guidelines. Based on this, the following hypotheses are proposed:

H4: In medical research activities, higher levels of GenAI-perceived risk are significantly associated with lower awareness of research integrity among medical researchers.

2.2.2. Technical trust (TT)

Previous research has identified technical trust as a critical determinant of artificial intelligence adoption (Cheng et al. 2022; Tanantong, Nantajeewarawat, and Thiemjarus 2016). In social science research, technical trust has been applied as a key variable in studies on smart home healthcare services and autonomous driving technologies (Shugang et al. 2023; Wieland, Michael, and Marek 2025). In this study, technical trust is defined as medical researchers' perceptions of GenAI's understandability, reliability, and interpretability. It captures researchers' confidence in the accuracy, transparency, and verifiability of AI-generated outputs. When researchers hold a high level of technical trust in GenAI, they are more likely to explicitly disclose its use, rigorously validate its outputs, and adhere to established research-integrity standards. Such trust encourages rational engagement with GenAI – valuing not only efficiency but also the authenticity and traceability of research data – thereby reinforcing integrity awareness. Conversely, insufficient trust may lead researchers to conceal AI usage to avoid scrutiny, undermining their commitment to ethical research practices. Based on this reasoning, we propose:

H5: In medical research activities, higher technical trust in GenAI is significantly associated with greater awareness of research integrity among medical researchers.

2.2.3. *GenAI anxiety (GenAI-AY)*

Previous research has applied the concept of AI anxiety to areas such as unemployment-related anxiety and knowledge payment systems, with some studies indicating that cognitive characteristics may alleviate anxiety associated with artificial intelligence (Jinsong, Miao, and Jinhua 2025; Mehmet, Ç, and Muhammet Faruk 2024; Xintong and Yipeng 2025). GenAI anxiety refers to the apprehension and unease that medical researchers may experience when interacting with GenAI tools in the course of their scientific work (Budhathoki et al. 2024). Such anxiety can undermine researchers' focus on integrity norms, diminish their willingness to adhere to ethical standards, and impair their judgment regarding research ethics. Moreover, heightened GenAI anxiety may prompt avoidance or maladaptive coping strategies, increasing the risk of engaging in misconduct such as plagiarism or data fabrication. Based on this, the following hypotheses are proposed:

H7: In medical research activities, higher levels of GenAI anxiety are significantly associated with lower awareness of research integrity among medical researchers.

H9: In medical research activities, higher levels of GenAI anxiety are significantly associated with reduced willingness to adhere to research integrity.

2.3. *Medical research integrity awareness (MRIA)*

Research on the characteristics of retracted papers in the medical field has primarily focused on factors such as the reasons for retraction, time to retraction, journal impact factor, number of retracted articles, authorship patterns, involved journals, and disciplinary domains (Kocyigit, Akyol, et al. 2023). Common reasons for retraction include data fabrication or falsification (Deng and Wu 2021; Ren, Jin, and Weng 2019), duplicate publication (Jin, Liao, and Zhou 2023), and violations of ethical standards (Sun, Wang, and Ding 2023). For instance, 41.8% of retracted medical papers were withdrawn due to fraudulent peer review practices (Kocyigit, Zhaksylyk, et al. 2023). Additional causes include third-party authorship or submission services, commercialization of manuscripts (Y. Jiang, Liu, and Sun 2022), fabrication of the research process, and manipulation or falsification of research data (Y. Jiang, Liu, and Sun 2022; Kaibing, Xu, and Jue 2023).

Some studies have conducted in-depth investigations into prominent academic misconduct events. Notable examples include repeated publications by a single research group at a major university (Cui, Zhao, and Zheng 2018), the high-profile “Chen Jin Incident” in China (M. Jiang, Jiang, and Duan 2013), the

mass retraction of 117 papers authored by Chinese scholars by an international publishing group in 2015 (Li and Wang 2019), and the retraction of 107 papers by the journal *Tumor Biology* in 2017 (X. Li, Chen, and Guan 2022). These events have had a profound negative impact on the reputation of the associated institutions and disciplines, and have significantly damaged the international credibility of academic research originating from the country (Wu et al. 2022). Among the various forms of misconduct, violations of ethical norms – particularly in clinical and animal research – have emerged as especially prominent in the medical sciences (Ma 2022).

In this study, medical research integrity awareness refers to the intrinsic disposition or attitudinal orientation of medical researchers toward adhering to ethical research norms and preserving the authenticity and objectivity of scientific inquiry when utilizing GenAI-assisted methods. This construct reflects the degree to which researchers recognize and value the principles of research integrity and constitutes a foundational element of their ethical cognition. As a key internal cognitive driver, research integrity awareness fosters a proactive commitment to normative conduct and reduces the likelihood of research misconduct. Based on this, the following hypotheses are proposed:

H8: In medical research activities, higher awareness of research integrity significantly promotes compliance with research integrity behaviors among medical researchers.

Based on the above hypotheses, the research hypothesis diagram is as follows (Figure 2). Applications of UTAUT in the context of GenAI in medical research remain limited. To address this gap, the present study extends the original UTAUT framework by incorporating variables relevant to the unique characteristics of GenAI within medical research workflows. Specifically, we introduce GenAI perceived risk, technology trust, and GenAI anxiety to capture the psychological and ethical challenges that such emerging technologies may pose. This adapted model is employed to analyze the key factors influencing medical researchers' awareness and behavior concerning research integrity.

3. Research method

3.1. Research design, Methodology, and data collection technique

This study employed a quantitative approach to examine causal relationships between GenAI and medical research integrity. A cross-sectional survey design was adopted, targeting individuals involved in medical research across

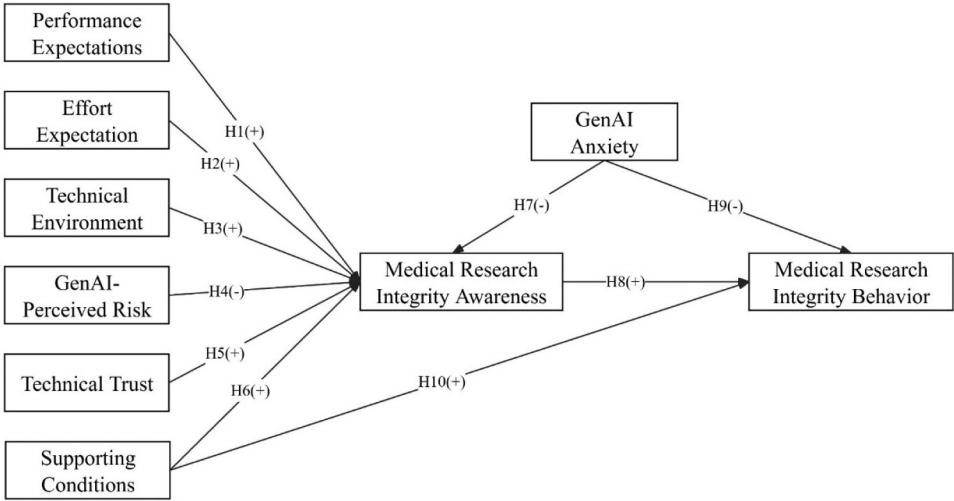


Figure 2. Conceptual framework: GenAI’s influence on medical research integrity.

China, including students, faculty members, and administrative managers in medical institutions.

Data collection occurred between 30 November 2024, and 20 January 2025. Eight hundred survey invitations containing a URL link were distributed to researchers randomly selected from various medical institutions. To increase the questionnaire completion rate, brief motivational messages were embedded at key points (e.g., “One-third of the questionnaire has been completed – thank you for your participation!”; “Two-thirds of the questionnaire has been completed – your effort is greatly appreciated!”; “Thank you for completing the questionnaire – your contribution is outstanding!”). In total, 718 valid responses were obtained, resulting in a response rate of approximately 89.8%.

Prior to full-scale distribution, a pilot study was conducted to assess and refine the questionnaire. Based on feedback from the pretest, several items were modified to enhance clarity and to ensure the reliability and validity of the measurement model examining GenAI’s impact on medical research integrity.

3.2. Measurements and analysis tool

The questionnaire comprised 69 items designed to measure nine constructs: performance expectancy, effort expectancy, technical environment, GenAI-perceived risk, technical trust, supporting conditions, GenAI anxiety, awareness of medical research integrity, and research integrity behavior. All items were adapted from previously validated instruments and reworded to reflect the context of GenAI’s influence on medical research integrity. [Table 1](#)

Table 1. Compilation of constructs sources for the present study.

NO.	Constructs	Original References
1	Performance expectations	Venkatesh et al. (2003), Menon and Shilpa (2023), Gong, Deng, and Cao (2022), S. Guo and Zhang (2024), Tian et al. (2024)
2	Effort expectation	Venkatesh et al. (2003), Muriithi, Horner, and Pemberton (2016), S. Guo and Zhang (2024), Duan, Zhang, and Wang (2023)
3	Technical environment	Venkatesh et al. (2003), Fu et al. (2020), Budhathoki et al. (2024)
4	GenAI-Perceived risk	Chen, Fan, and Azam (2024)
5	Technical trust	Chen, Fan, and Azam (2024)
6	Supporting conditions	Venkatesh et al. (2003), Menon and Shilpa (2023), Andrews, Ward, and Yoon (2021), S. Guo, Zhang, and Su (2024)
7	GenAI-Anxiety	Budhathoki et al. (2024)
8	Medical research integrity awareness	Venkatesh et al. (2003)
9	Medical research integrity behavior	Venkatesh et al. (2003)

summarizes the measurement sources for each construct. Specifically, the scales for medical research integrity awareness and medical research integrity behavior were derived from: the “Notice on Medical Research Integrity and Related Behavioral Norms” (National Health Commission of the PRC, 2021) (China N H C o t P s R o 2021); the “Boundary Guidelines for the Use of AIGC in Academic Publishing” (Institute of Scientific and Technical Information of China, 2023) (China I o S a T I o 2024); and the “Manual of Research Integrity Norms” (National Natural Science Foundation of China, 2023) (China N N s F o 2023). Respondents rated all items on a seven-point Likert scale ranging from 1 (“strongly disagree”) to 7 (“strongly agree”). Data were analyzed using covariance-based structural equation modeling (SEM) to assess both the measurement model and the hypothesized structural relationships. The constructs and their corresponding items are shown in [Appendix A](#).

4. Data analysis and findings

4.1. Sample profile

4.1.1. Demographic characteristics of the respondents

Table 2 presents the demographic characteristics of the 718 respondents. The sample comprised 383 males (53.3%) and 335 females (46.7%), indicating a balanced gender distribution. Age was distributed as follows: 18–25 years (43.5%), 26–30 years (29.7%), 31–40 years (11.1%), 41–50 years (7.3%), 51–60 years (4.3%), and over 60 years (4.0%). Regarding educational background, 41.1% held a master’s degree, 40.7% a bachelor’s degree, 11.3% a doctoral degree, and 7.0% an associate degree.

Table 2. Demographic characteristics of the respondents.

Measure	Option	N (n = 718)	n%
Gender	Female	335	46.7
	Male	383	53.3
Age (years)	18–25	312	43.5
	26–30	213	29.7
	31–40	80	11.1
	41–50	53	7.4
	51–60	31	4.3
	Over 60	29	4.0
Educational Background	Associate Degree	50	7.0
	Bachelor's Degree	292	40.7
	Master's Degree	295	41.1
	Doctoral Degree	81	11.3
	Other	0	0
Occupation	University faculty	23	3.2
	University administrator	36	5.0
	Hospital administrator	31	4.3
	Medical professional	35	4.9
	Hospital support staff	34	4.7
	Student	554	77.2
	Other	5	0.7
Academic Discipline	Public Health and Preventive Medicine	110	15.3
	Basic Medical Sciences	155	21.6
	Clinical Medicine	83	11.6
	Nursing	107	14.9
	Health Management and Administration	120	16.7
	Pharmacy	130	18.1
	Other	13	1.8
Academic Title	Junior Title	170	23.7
	Intermediate Title	62	8.6
	Associate Professor	8	1.1
	Professor	6	0.8
	None	458	63.8
	Other	14	2.0

In terms of occupation, 77.2% were students, 13.9% worked in medical institutions (e.g., hospital administrator, medical professional, hospital support staff), and 8.2% were faculty or administrators at medical colleges and universities. The most academic discipline were basic medical sciences (21.6%), pharmacy (18.1%), Health management and administration (16.7%), public health and preventive medicine (15.3%), nursing (14.9%), and clinical medicine (11.6%), with other disciplines comprising 1.8% of the sample. Finally, professional titles were reported as follows: no title (63.8%), junior (23.7%), intermediate (8.6%), associate professor (1.1%), professor (0.8%), and other titles (2.0%).

4.1.2. Analysis of awareness levels of GenAI and medical research integrity

As shown in Table 3, in response to “How familiar are you with GenAI?,” 5.6% of respondents reported being “Extremely familiar,” and 23.3% reported “General understanding,” totaling 29.4%. This indicates

Table 3. Perceptions of GenAI and research integrity in medicine.

Items	Option	Frequency	Percentage (%)
How familiar are you with GenAI?	Extremely familiar	40	5.6
	Very familiar	171	23.8
	Moderately familiar	151	21.0
	General understanding	194	27.0
	Slightly unfamiliar	104	14.5
	Quite unfamiliar	33	4.6
How familiar are you with Medical Research Integrity?	Completely unfamiliar	25	3.5
	Extremely familiar	44	6.1
	Very familiar	78	10.7
	Moderately familiar	154	21.5
	General understanding	273	38.0
	Slightly unfamiliar	101	14.0
	Quite unfamiliar	38	5.3
	Completely unfamiliar	30	4.1

that nearly 30% of medical researchers possess a relatively deep understanding of GenAI. Additionally, 21.0% identified as “Moderately familiar,” and 27.0% as “General understanding” (48.0% combined), suggesting that about half have some – but not comprehensive – knowledge of GenAI. Meanwhile, 14.5% were “Slightly unfamiliar,” 4.6% “Quite unfamiliar,” and 3.5% “Completely unfamiliar” (22.6% combined), indicating that a minority have little or no familiarity. Overall, most respondents fall in the mid-range categories, with relatively few at the extremes.

In response to “How familiar are you with Medical Research Integrity?” (Table 3), the largest group (38.0%) selected “General understanding,” whereas only 6.1% indicated they were “Extremely familiar,” suggesting that most researchers have a basic grasp of research integrity, but few possess in-depth knowledge. The combined proportion of “Extremely familiar” and “Very familiar” was 17.0%, confirming that only a minority have a thorough understanding. Meanwhile, 59.5% reported either “Moderately familiar” or “General understanding,” placing them at an intermediate awareness level. Finally, 23.5% were “Slightly unfamiliar,” “Quite unfamiliar,” or “Completely unfamiliar,” highlighting significant gaps in understanding. Excluding these less familiar categories, the distribution of awareness resembles a pyramid: those at “Moderately familiar” and “General understanding” form the base with basic knowledge; those at “Very familiar” occupy the middle tier with more substantial understanding; and the small proportion of “Extremely familiar” constitutes the apex, representing the most comprehensive knowledge of medical research integrity. (3) Analysis of GenAI Usage

Table 4 presents the usage rates of various GenAI tools. “Wenxin Yiyan” leads with 69.4%, indicating its widespread adoption among medical researchers. “ChatGPT” follows at 57.4%, reflecting its global prominence.

Table 4. GenAI use cases in medical research.

Items	Option	Frequency	Percentage (%)
Which GenAI tools have you used? (Multiple selections allowed)	ChatGPT	264	57.4
	Xunfei Xinghuo	47	10.2
	Wenxin Yiyan	319	69.4
	Tongyi Qianwen	147	32.0
	DouBao	244	53.0
	Tencent Yuanbao	155	33.7
	Tiangong	105	22.8
	Kimi	139	30.2
	Other	36	7.8
How have you used GenAI in medical research activities? (Multiple selections allowed)	Generating images, videos, audio, or datasets	78	10.9
	Manuscript writing, translation, or editing	513	71.5
	Literature retrieval, summarization, and refinement	374	52.1
	Query resolution	225	31.3
	Code generation	473	65.9
	Presentation design	299	41.6
	Statistical analysis	268	37.3
	Chart creation	221	30.8
	Citation organization	245	34.1
	Other	36	5.0

“Tongyi Qianwen,” “Tencent Yuanbao,” and “Kimi” show moderate usage rates of 32.0%, 33.7%, and 30.2%, respectively. “Xunfei Xinghuo” and “Tiangong” have lower adoption rates of 10.2% and 22.8%, respectively. Tools not listed in the options account for 7.8% of usage, suggesting limited use of alternative GenAI platforms.

Regarding application scenarios in medical research, “Manuscript writing, translation, or editing” is the most common use at 71.5%, demonstrating GenAI’s significant role in academic writing. “Code generation” follows at 65.9%, indicating extensive application in programming and data analysis. “Literature retrieval, summarization, and refinement” is used by 52.1% of respondents, highlighting GenAI’s role in literature management. “Presentation design,” “Statistical analysis,” and “Citation organization” have moderate usage rates of 41.6%, 37.3%, and 34.1%, respectively. “Query resolution” and “Chart creation” are used by 31.3% and 30.8%, respectively. “Generating images, videos, audio, or datasets” is the least common scenario at 10.9%, while other uses account for 5.0%. Overall, Wenxin Yiyan and ChatGPT are the most frequently utilized GenAI tools. GenAI’s applications in medical research are diverse, encompassing academic writing, programming, literature management, statistical analysis, and more.

4.2. Measurement model

This study assessed the reliability and validity of constructs in the model examining GenAI's impact on medical research integrity. Cronbach's α for all constructs ranged from 0.80 to 0.98, well above the 0.70 benchmark, indicating strong internal consistency. All factor loadings exceeded 0.70, confirming item reliability. Composite reliability (CR) for each construct was above 0.70, and average variance extracted (AVE) exceeded 0.50, demonstrating satisfactory convergent validity. Table 5 summarizes these measurement-model indices. Discriminant validity was also supported: as shown in Table 6, the square root of each construct's AVE surpassed its correlations with other constructs, confirming that each latent variable is distinct. The constructs and their corresponding items are shown in Appendix A.

4.3. Structural model

As shown in Table 7, the model exhibits excellent fit. The chi-square-to-df ratio (CMIN/DF) is 1.143. Fit indices exceed conventional thresholds: NFI = 0.964, IFI = 0.995, TLI = 0.995, and CFI = 0.995 (all > 0.90). The GFI of 0.933 likewise surpasses 0.90, and the RMSEA is 0.014 (< 0.05). Collectively, these indices indicate that the hypothesized model fits the data well and effectively addresses the research questions.

Table 8 presents standardized path coefficients. Performance expectancy ($\beta = 0.125$, $p < 0.001$), effort expectancy ($\beta = 0.084$, $p = 0.018$), technical environment ($\beta = 0.151$, $p < 0.001$), technical trust ($\beta = 0.228$, $p < 0.001$), and supporting conditions ($\beta = 0.118$, $p = 0.002$) all show significant positive associations with integrity awareness. In contrast, GenAI-perceived risk ($\beta = -0.085$, $p = 0.020$) and GenAI anxiety ($\beta = -0.109$, $p = 0.002$) are negatively related to integrity awareness. Furthermore, integrity awareness ($\beta = 0.265$, $p < 0.001$) and supporting conditions ($\beta = 0.233$, $p < 0.001$) positively predict integrity behavior, whereas GenAI anxiety negatively predicts integrity behavior ($\beta = -0.088$, $p = 0.021$). These results support Hypotheses 1–10.

5. Discussion and conclusions

5.1. Discussion

5.1.1. Technical drivers enhancing integrity awareness

This study demonstrates that performance expectancy, effort expectancy, technical environment, trust in technology, and supporting conditions are each positively associated with medical research integrity awareness, which exhibits a degree of concordance with the conclusions by Bouteraa (Mohamed et al. 2024). These findings reveal the underlying mechanisms through which technical factors influence researchers' perceptions of

Table 5. Measurement model results.

Construct	Item	Loading	Cronbach's α of item	Cronbach's α of construct	AVE	CR
Performance expectations(PE)	PE1	0.891	0.949	0.957	0.789	0.957
	PE2	0.889	0.949			
	PE3	0.886	0.949			
	PE4	0.89	0.949			
	PE5	0.888	0.949			
	PE6	0.886	0.949			
Effort expectation(EE)	EE1	0.885	0.937	0.949	0.789	0.949
	EE2	0.893	0.938			
	EE3	0.885	0.938			
	EE4	0.886	0.936			
	EE5	0.892	0.938			
Technical environment(TE)	TE1	0.897	0.885	0.916	0.785	0.916
	TE2	0.886	0.881			
	TE3	0.874	0.870			
GenAI-Perceived risk(GenAI-PR)	PR1	0.885	0.882	0.922	0.798	0.922
	PR2	0.892	0.888			
	PR3	0.902	0.892			
Technical trust(TT)	TT1	0.875	0.853	0.892	0.734	0.892
	TT2	0.852	0.851			
	TT3	0.843	0.834			
Supporting conditions(SC)	SC1	0.874	0.869	0.913	0.777	0.913
	SC2	0.883	0.876			
	SC3	0.888	0.879			
GenAI-Anxiety (GenAI-AY)	AY1	0.872	0.875	0.912	0.776	0.912
	AY2	0.893	0.864			
	AY3	0.877	0.879			
Medical research integrity awareness (MRIA)	MRIA1	0.881	0.970	0.973	0.750	0.973
	MRIA2	0.885	0.970			
	MRIA3	0.836	0.971			
	MRIA4	0.887	0.970			
	MRIA5	0.859	0.970			
	MRIA6	0.902	0.969			
	MRIA7	0.894	0.970			
	MRIA8	0.835	0.971			
	MRIA9	0.894	0.970			
	MRIA10	0.865	0.970			
	MRIA11	0.851	0.971			
	MRIA12	0.793	0.972			
Medical research integrity behavior (MRIB)	MRIB 1	0.846	0.972	0.974	0.741	0.974
	MRIB 2	0.856	0.972			
	MRIB3	0.876	0.971			
	MRIB4	0.877	0.971			
	MRIB5	0.880	0.971			
	MRIB6	0.871	0.971			
	MRIB7	0.880	0.971			
	MRIB8	0.866	0.971			
	MRIB9	0.827	0.972			
	MRIB10	0.873	0.971			
	MRIB11	0.862	0.972			
	MRIB12	0.818	0.972			
	MRIB13	0.855	0.972			

Table 6. Discriminant validity.

	MRIB	MRIA	GenAI-AY	SC	TT	GenAI-PR	TE	EE	PE
MRIB	0.861								
MRIA	0.37	0.866							
GenAI-AY	−0.218	−0.308	0.881						
SC	0.338	0.356	−0.235	0.882					
TT	0.416	0.452	−0.299	0.331	0.857				
GenAI-PR	−0.291	−0.304	0.208	−0.277	−0.338	0.893			
TE	0.39	0.408	−0.234	0.353	0.409	−0.296	0.886		
EE	0.251	0.278	−0.253	0.317	0.242	−0.163	0.257	0.888	
PE	0.361	0.353	−0.216	0.25	0.348	−0.242	0.411	0.173	0.888

The bold diagonal elements (from the top-left to the bottom-right) indicate the square roots of the Average Variance Extracted (AVE) values.

Table 7. Table of structural validity indices for the questionnaire.

Evaluation Index	Model Output	Recommended Threshold	Model Fit
CMIN/DF	1.143	<2	Better
NFI	0.964	≥0.90	Better
IFI	0.995	>0.90	Better
TLI	0.995	≥0.90	Better
CFI	0.995	≥0.90	Better
GFI	0.933	≥0.90	Better
RMSEA	0.014	<0.05	Better

Table 8. Structural model results.

Hypotheses	Hypotheses	β	Std error	C.R.	P-value	Supported?
H1	PE→MRIA	0.125	0.035	3.388	0.000	Yes
H2	EE→MRIA	0.084	0.034	2.365	0.018	Yes
H3	TE→MRIA	0.151	0.037	3.726	0.000	Yes
H4	GenAI-PR→MRIA	−0.085	0.034	−2.334	0.02	Yes
H5	TT→MRIA	0.228	0.044	5.555	0.000	Yes
H6	SC→MRIA	0.118	0.037	3.067	0.002	Yes
H7	GenAI-AY→MRIA	−0.109	0.039	−3.029	0.002	Yes
H8	MRIA→MRIB	0.265	0.036	6.732	0.000	Yes
H9	GenAI-AY→MRIB	−0.088	0.037	−2.302	0.021	Yes
H10	SC→MRIB	0.233	0.035	5.89	0.000	Yes

integrity. Specifically, performance expectancy leads medical researchers to view GenAI as a valuable tool for improving the quality and efficiency of research. In specialized domains – such as optimizing clinical diagnosis and treatment pathways or exploring disease-drug mechanisms – the credibility of GenAI-assisted outcomes depends heavily on the authenticity of research data and the standardization of analytical procedures, thereby reinforcing integrity awareness. The positive impact of effort expectancy suggests that when researchers perceive GenAI as easy to learn and use, they are more inclined to incorporate it into their workflows in a manner consistent with ethical research norms. Improvements in the technical environment provide clear operational guidelines for GenAI-supported research, helping to

prevent unintentional misconduct arising from regulatory ambiguity (Tanatorn, Ekawit, and Surapa 2015). Trust in technology further strengthens researchers' confidence in GenAI outputs – particularly in contexts such as drug trials, medical imaging, and multimodal data analysis – thus mitigating unethical practices like data manipulation or “result beautification.” Additionally, robust supporting conditions increase the accessibility and usability of GenAI, fostering an environment conducive to the development of integrity-conscious research behaviors. Moving forward, medical institutions are encouraged to cultivate a research integrity framework centered on the progression from “technology acceptance” to “trust building” to “awareness internalization,” supported through system optimization, training and education, and technical infrastructure, thereby reinforcing ethical standards in the medical research ecosystem.

5.1.2. Psychological and ethical pressures undermining integrity

This study further indicates that perceived risks and anxiety related to GenAI are negatively associated with medical research integrity awareness, which provides empirical support for the scholarly view that ethical considerations and academic integrity can shape perceptions of the use and utility of GenAI (Metwaly Ali Mohamed, Ahmed Maher Khafaga, and Abdelrahman Saber Abdelrahman 2025). These findings highlight how the combined psychological and ethical pressures introduced by emerging technologies can erode researchers' commitment to integrity. Medical research often involves extensive real-world data, particularly sensitive patient information. Researchers' concerns about the opacity and lack of transparency in GenAI's data processing mechanisms heighten perceived risks, which may lead some to limit data sharing or engage in nonstandard data manipulation – practices that compromise research integrity. Anxiety is especially prevalent among medical researchers operating in highly competitive environments, where professional advancement, project funding, and publication success are under constant pressure. Uncertainty about GenAI's effectiveness, fears of technological displacement, and unclear boundaries surrounding its lawful use contribute to hesitation, discomfort, and even avoidance in its application. Such emotional and cognitive stressors undermine researchers' ability to make objective judgments and adhere to ethical research practices. To address these challenges, medical research institutions should promote informed, rational, and compliant GenAI usage by implementing transparent risk communication strategies, clarifying usage norms, and offering targeted anxiety-reduction interventions. These efforts can help strengthen research integrity at its foundation.

5.1.3. From awareness to Action: Factors Influencing integrity behaviors

This study further demonstrates that awareness of medical research integrity and supporting conditions significantly promote integrity-related behaviors,

whereas GenAI-related anxiety exerts a detrimental effect. Integrity awareness, as an internalized driver of normative behavior, encourages medical researchers to emphasize verification and compliance when utilizing GenAI-generated content, thereby reducing misconduct such as plagiarism, data fabrication, and unauthorized reuse driven by overreliance on the technology. In contrast, GenAI-induced anxiety can impair researchers' judgment and diminish their willingness to adhere to ethical standards, especially under the pressure of performance evaluations and publication demands, increasing the risk of non-compliant use. To effectively standardize research behavior in the GenAI era, it is essential to strengthen integrity awareness, enhance institutional and technical support systems, and implement psychological interventions that mitigate anxiety and promote ethical decision-making.

5.2. Implications to theory and practice

The empirical findings derived from this study make substantive contributions to the advancement of theoretical frameworks and the enhancement of practical strategies. In terms of theoretical implications, this study grounded in the UTAUT model, integrates key psychological and cognitive variables related to the use of generative artificial intelligence – namely GenAI perceived risk, technology trust, and GenAI anxiety – to systematically examine the mechanisms that influence medical researchers' awareness of and adherence to research integrity. It expands the theoretical boundaries at the intersection of generative AI and research ethics. By incorporating research integrity as a behavioral outcome variable for the first time within the context of technology acceptance and use, this study highlights the critical role of both technical attitudes and emotional factors – such as AI-related anxiety – in shaping ethical research behavior. These findings enrich the theoretical framework of technology-related ethical behavior and provide valuable theoretical insights for constructing a model of technological ethics tailored to the medical research environment.

Concerning the practical implications, this study demonstrates that factors such as performance expectations, effort expectation, technical environment, and technology trust significantly enhance researchers' awareness of research integrity and their compliance behaviors. Accordingly, strategies should be adopted for different stakeholders in medical research integrity. For employees, it is essential to proactively develop a rational understanding of and competence in using GenAI tools. Researchers should be adept at leveraging GenAI to improve research efficiency while maintaining a strong sense of responsibility for the content it generates. Moreover, they must remain vigilant against improper use stemming from technical anxiety or cognitive biases, strengthen self-discipline, and refrain from using AI-generated content for publication without adequate review, thus upholding the ethical standards of scientific research. For managers, attention should be paid to fluctuations in researchers'

trust in technology and anxiety during the use of GenAI. Effective psychological support and technical training mechanisms should be established. On the one hand, organizing targeted training sessions and experience-sharing initiatives can enhance researchers' proficiency with and confidence in AI tools. On the other hand, risk identification and feedback channels should be implemented to help researchers recognize and mitigate potential ethical risks. In particular, managers must clearly define the boundary between AI-assisted and human-generated content and develop usage guidelines and disclosure protocols to reduce the risk of research misconduct.

5.3. Limitations and future directions

This study elucidates the mechanism through which GenAI influences medical research integrity, offering a theoretical foundation for regulating AI applications and safeguarding research authenticity. Nonetheless, several limitations and opportunities for future research remain. First, our analysis did not differentiate among specific medical professions or disciplines. Future studies should examine how GenAI's impact on research integrity varies across diverse research paradigms and specialty areas. Second, we did not explore the co-evolutionary dynamics between technological perceptions and research integrity within interdisciplinary contexts. Finally, our sample was limited by the accessibility of medical faculty, hospital researchers, and administrative personnel, which may introduce bias and constrain the generalizability of our findings. Future research could examine how evaluation systems across various disciplines address the risks associated with technological misuse from the perspectives of science, engineering, the humanities, and the social sciences. Additionally, cross-cultural and cross-sectional studies involving medical researchers from diverse cultural and social backgrounds could be conducted to explore how national research integrity policies influence the role of generative artificial intelligence in advancing scientific development.

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Ethical Approval

This study was reviewed and approved by the Ethics Committee of the School of Life Sciences at Central South University (Approval No. 2024-1-6-18).

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Appendix A. Constructs and their corresponding items

Construct	Item
Performance expectations	I believe that the use of GenAI can ...
	PE1 meet the needs of medical research.
	PE2 help solve problems in medical science.
	PE3 provide additional benefits and value.
	PE4 enhance the comprehensiveness of scientific research.
	PE5 improve the accuracy of research findings.
Effort expectation	PE6 increase the timeliness of scientific research.
	EE1 I believe that GenAI is easy to access.
	EE2 I believe that my understanding of GenAI is influenced by my level of domain knowledge.
	EE3 I believe that my use of GenAI is influenced by my prior technical experience.
	EE4 I believe that GenAI is easy to operate and learn.
Technical environment	EE5 I believe that using GenAI enables me to quickly teach others how to use it.
	TE1 I believe that the use of GenAI tools is influenced by national or regional policies.
	TE2 I believe that the use of GenAI tools is influenced by the regulations of publishers and other copyright-related organizations.
GenAI-Perceived risk	TE3 I believe that the use of GenAI tools is influenced by the institutional framework of the researcher's affiliated organization.
	PR1 I believe that using GenAI tools may lead to the leakage of personal privacy.
	PR2 I believe that using GenAI tools may produce incorrect outcomes.
Technical trust	PR3 I believe that using GenAI tools may violate ethical standards.
	TT1 I believe that the content generated by GenAI tools is understandable.
	TT2 I believe that the content generated by GenAI tools is trustworthy.
Supporting conditions	TT3 I believe that the content generated by GenAI tools is interpretable.
	SC1 I believe that using GenAI tools requires reliable hardware infrastructure.
	SC2 I believe that using GenAI tools requires a user-friendly interactive interface.
GenAI-Anxiety	SC3 I believe that using GenAI tools requires a secure and stable network environment.
	AY1 I feel adaptive anxiety when using GenAI tools.
	AY2 I feel dependency-related anxiety when using GenAI tools.
Medical research integrity awareness	AY3 I feel career-related anxiety when using GenAI tools.
	I know that GenAI can assist medical research activities during ...
	MRIA1 the literature search and organization stage.
	MRIA2 the secondary creation, chart type recommendation, and assisted chart production stage.
	MRIA3 the code writing, debugging, and error checking stage.

(Continued)

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Construct	Item
Medical research integrity behavior	MRIA4 the screening and recommending of statistical methods, experimental methods, and survey methods stage.
	MRIA5 the reference formatting stage.
	MRIA6 the research design and data analysis stage.
	MRIA7 the raw data collection stage.
	MRIA8 the creation of result images and key illustrations stage.
	MRIA9 the manuscript writing stage.
	MRIA10 the language polishing and translation stage.
	MRIA11 the defense, review, and inspection stage.
	MRIA12 the handling of confidential content stage.
	Using GenAI in medical research activities ...
	MRIB 1 I verify the accuracy and authenticity of reference materials.
	MRIB 2 I validate the results of statistical analyses.
	MRIB3 I ensure that experiment-generated images originate from genuine experimental studies.
	MRIB4 I personally complete the writing of text.
	MRIB5 I cite and annotate data sources, tools, acquisition methods, and processing procedures.
	MRIB6 I provide full and accurate disclosure and declaration of GenAI usage.
	MRIB7 I adhere to national, institutional, and journal research integrity standards.
	MRIB8 I conduct self-inspection of content generated by GenAI.
	MRIB9 I maintain transparency and accountability by understanding and clearly disclosing the use of GenAI.
	MRIB10 I ensure quality and integrity by indicating the use of GenAI through identifiers or feedback.
	MRIB11 I respect privacy and data protection to uphold privacy and security.
	MRIB12 I promote fairness by avoiding bias and reducing cultural or linguistic inequities.
	MRIB13 I support sustainability by avoiding overreliance on data that may be temporarily or permanently unavailable, ensuring data traceability.